Global CO_2 Emissions: Trends, Regional Shifts, and Per Capita Inequalities

A data-driven overview from 1750–2024

Data Source: Global Carbon Budget 2025, NASA

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial Levels

429

parts per million (ppm)

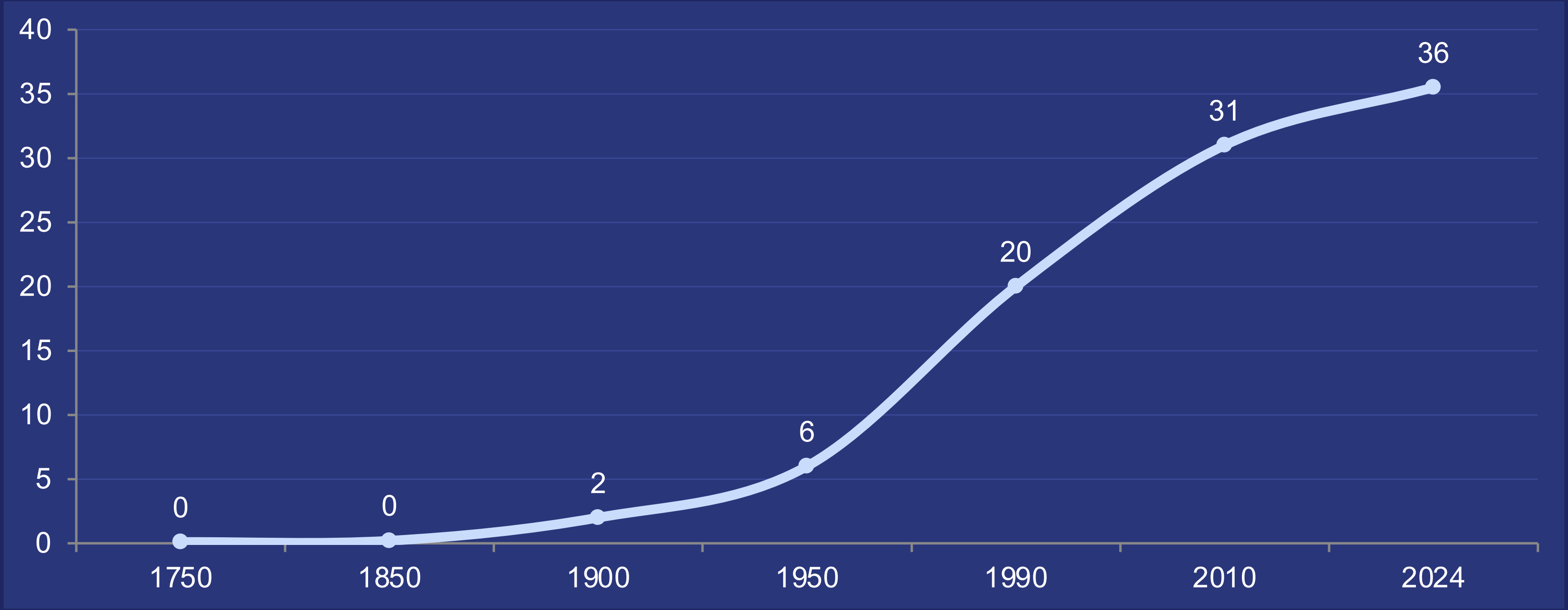
Latest measurement from NOAA's Mauna Loa Observatory (Feb 2026)

Pre-Industrial Baseline: ~280 ppm

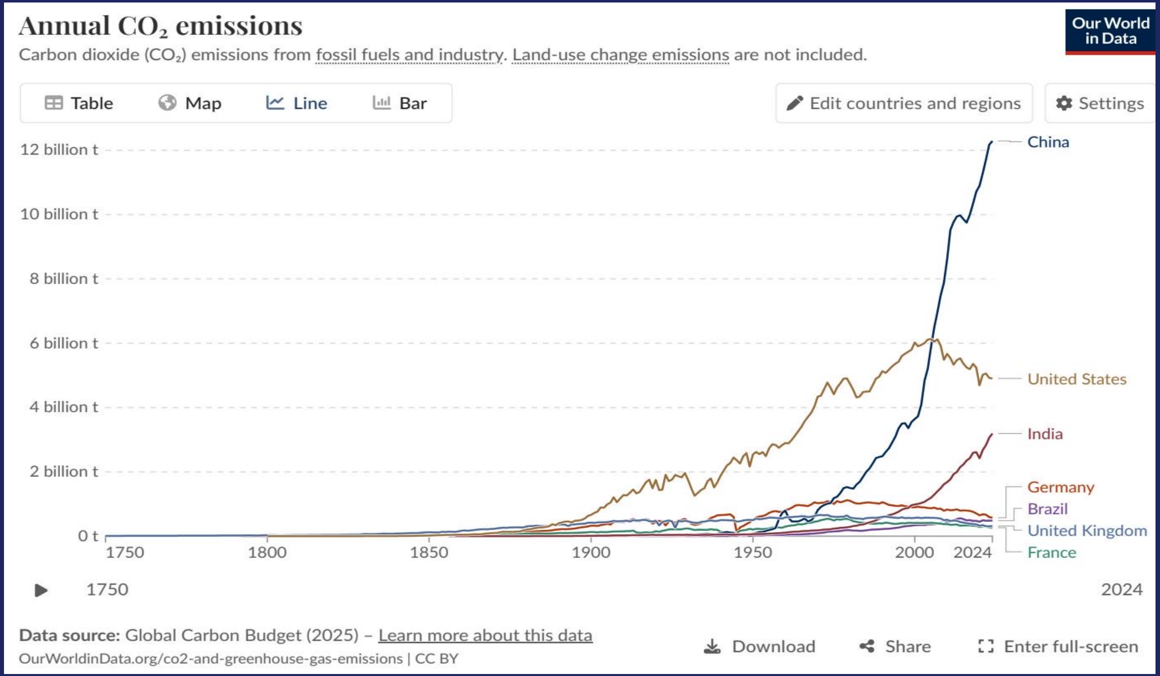
Increase: >50% growth

The current level is 1.5× greater than the natural increase at the end of the last ice age—a shift that previously took thousands of years, happening now in just two centuries.

Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



China Now Emits Over 12 Billion Tonnes — More Than Double the United States



Top Emitters

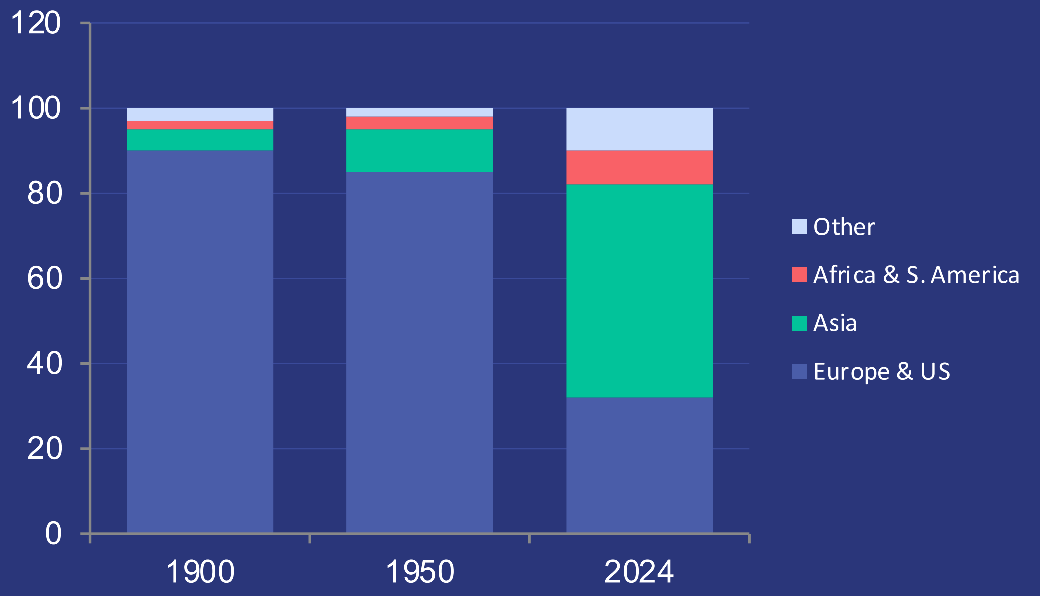
China
~12.5 Bt (27%)

United States
~5.0 Bt (14%) - Declining

India
~3.0 Bt (8%) - Rising

EU-27
~2.7 Bt (7%) - Declining

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



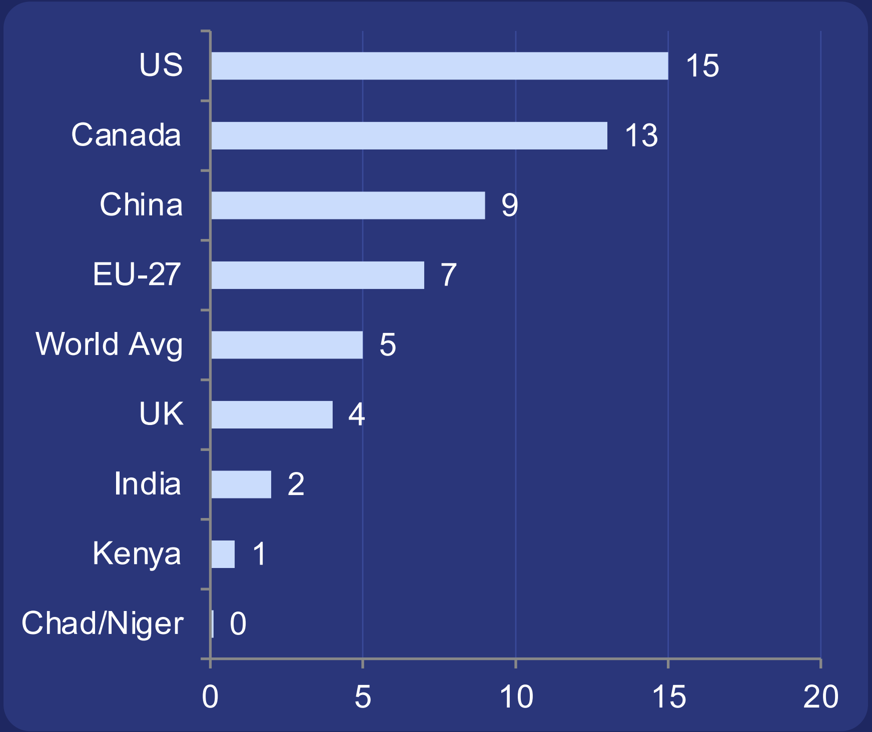
The Geographic Shift

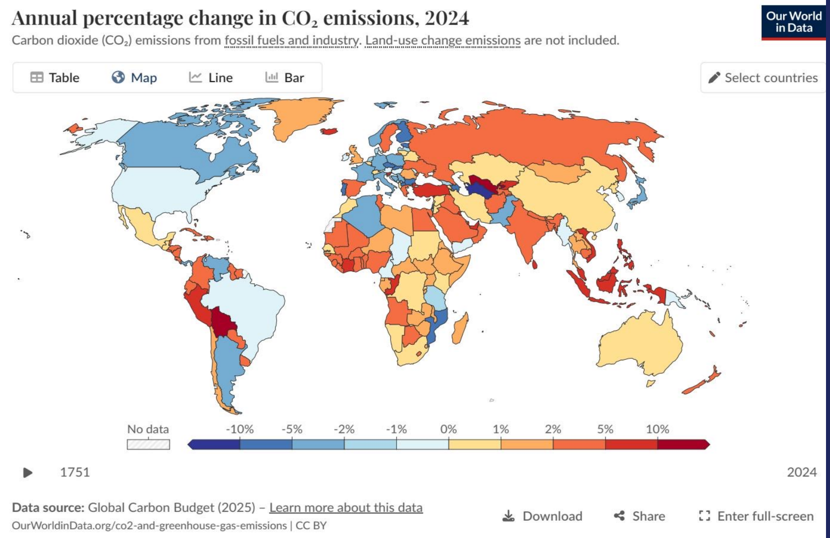
1900: Europe and the US accounted for >90% of global emissions.

Today: Asia (home to 60% of world pop) produces roughly half.

Africa & South America: Only 3-4% each, comparable to intl. aviation & shipping.

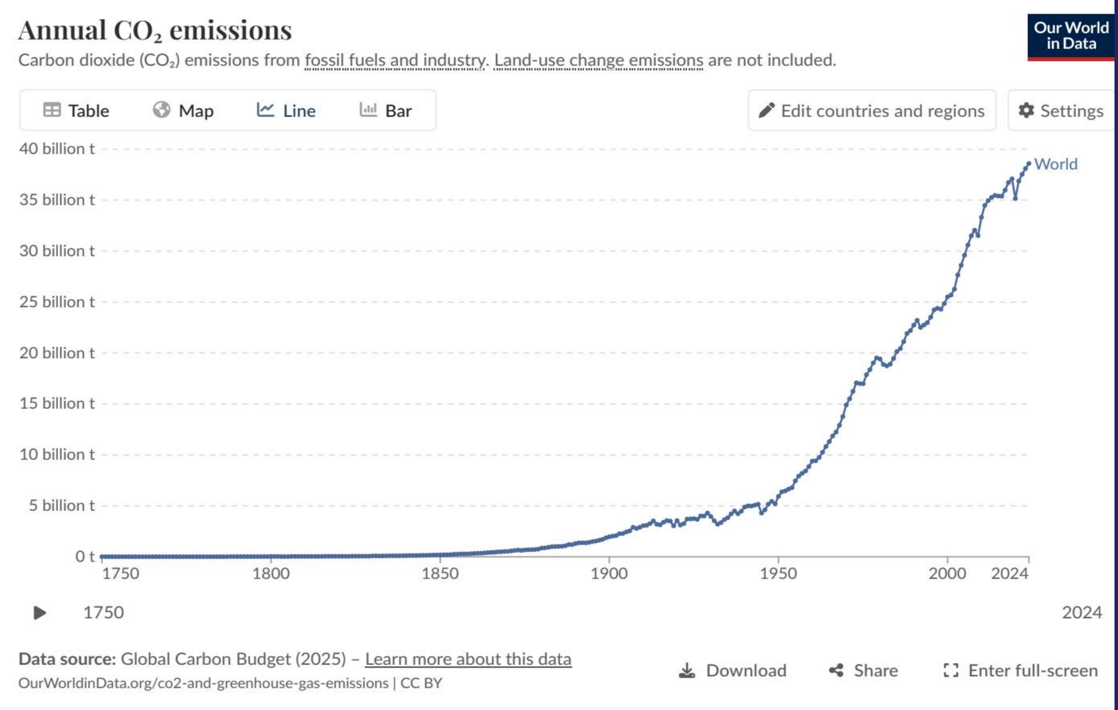
Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity





The average American emits in under 2 days what an average person in Chad emits in an entire year.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase



Annual Change 2024

Declining (-1% to -5%)

United States and much of Europe continue downward trends.

Increasing (+2% to >10%)

Russia, parts of SE Asia, and several African nations saw significant increases.

South America

Mixed picture: some sharp increases alongside steady declines.

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards differ significantly based on their primary energy sources.

Country	Per Capita CO_2	Primary Energy Driver
France	~4 t	High Nuclear & Renewables
United Kingdom	~4 t	Wind & Renewables
Germany	~6 t	Higher Coal & Fossil Share
Netherlands	~7 t	Higher Natural Gas

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

- Global emissions exceed 35 Bt and have not yet peaked.
- China alone emits >25% of the global total, double the US.
- Per capita inequalities remain extreme (a 150× gap between highest and lowest).
- US and Europe are declining, but their historical cumulative responsibility remains large.
- Policy and energy mix choices determine per capita outcomes, not just economic prosperity.